Identifying the 2024–2026 Cocoa Crisis as a Structural Break in Soft-Commodity Correlation Dynamics

A Random-Matrix-Cleaned DCC-GARCH Approach

FNCE40003 Numerical Techniques in Finance Major Assignment May 2026

1. Introduction

Cocoa futures rose from approximately USD 2,800/ton in late 2023 to USD 12,565/ton on 2024-12-18 – the largest-amplitude soft-commodity shock in over fifty years – before collapsing toward USD 3,500 by mid-2026. The drivers are well documented [1]: a 2023 El Niño, the spread of cacao swollen-shoot virus in West Africa, and divergent policy responses in Ghana and Côte d'Ivoire (together responsible for roughly 60% of global cocoa). We ask whether the crisis is visible as a structural break in the dependence structure of the broader soft-commodity complex, and how it reorganises that structure.

The paper makes three contributions. First, we apply the random-matrix cleaning of [2] to the DCC-GARCH of [3] on a 25-asset cocoa-adjacent universe, replacing the sample-correlation DCC target with a Marchenko–Pastur-cleaned long-run target, and document that at this universe size ($N = 25$, $Q \approx 90$) the cleaned target is rejected by every standard criterion. Second, we pre-register tests on the rolling supra-MP count and on the eigenvector composition, and report verdicts under BIC-selected Bai–Perron and a stationary block bootstrap. Third, we show that the cocoa NY–London basis *re-couples* during the crisis, consistent with a global supply shock affecting both contracts rather than the regional decoupling we anticipated.

2. Data

Block	Series
Cocoa contracts (2)	COCOA_NY (ICE-US), COCOA_LDN (ICE-Europe)
Adjacent softs (5)	Coffee Arabica, Coffee Robusta, Sugar No. 11, Cotton, FCOJ
Grains, oilseeds, energy (5)	Corn, Soybeans, Wheat, Palm oil, WTI crude
Macro factors (3)	DXY, VIX, US 10Y yield
FX and sovereign credit (4)	BRL, VND, GHS, Ghana 10Y yield
Chocolate-maker equities (6)	HSY, MDLZ, NESN, SJM, SBUX, LISN

Table 1: The 25-asset cocoa-adjacent universe. Daily series, 2015-01-02 to 2026-05-21 (2,864 trading days on the cocoa NY calendar). All series from Refinitiv except VIX (FRED). Series held in native currency to avoid spurious FX-conversion co-movement [4]. Returns are log differences for prices and first differences (percentage points) for the two yield series.

Table 1 lists the universe. The aspect ratio is $Q = T/N \approx 115$ on the full sample; the complete-cases panel (post-Ghana entry, 2017-04 onward) gives 2,260 trading days and $Q \approx 90$, the sample on which the full-sample MP spectrum and the DCC are computed.

WTI's 2020-04-20 negative settle (-USD 37.63) is patched to NA on −04-20 and −04-21. Ghana 10-year yield data begins on 2017-04-20 (79% of trading days); rolling windows ending before April 2018 use $N = 24$, all subsequent windows use $N = 25$. Standardised residuals are winsorised at ± 5 to prevent any single observation from dominating a 252-day correlation matrix (the Bouchaud–Potters convention [5]; 150 of approximately 71,500 residual observations affected, 0.21%).

3. Methodology

3.1. Univariate volatility

The univariate filter must produce standardised residuals that are approximately white in level and square so that the rolling correlation matrices in §3.2 reflect contemporaneous dependence rather than residual ARCH or autocorrelation. We use GJR-GARCH$(1,1)$ [6] with [7] skewed Student-t (sstd) innovations as the primary specification:

$$h_{i,t} = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \, \mathbb{1}_{\{\varepsilon_{i,t-1}<0\}} + \beta_i h_{i,t-1}, \quad z_{i,t} = \varepsilon_{i,t} / \sqrt{h_{i,t}}. \tag{1}$$

The GJR specification extends [8] and [9] by adding an asymmetric-response term γ_i that absorbs the leverage effect on equity and FX series at the cost of one extra parameter. We prefer it to [10] EGARCH because (a) the parameters retain their familiar variance-equation interpretation, (b) the implied unconditional variance is finite under the explicit constraint $\alpha_i + \beta_i + \gamma_i/2 < 1$, and (c) numerical convergence is more robust across our heterogeneous universe; EGARCH is retained as Tier 3 in the fallback chain below. We prefer skewed-t to symmetric-t because cocoa returns and several FX/equity series exhibit material conditional skewness that a symmetric distribution forces into the variance equation. A likelihood-ratio test of sstd vs symmetric-t (one degree of freedom) rejects symmetry at the 5% level for eight of the 25 series (COTTON, WHEAT, WTI, BRL, GHS, HSY, SJM, VIX), with VIX the most extreme ($\chi^2 = 184.6$). sstd nests symmetric-t (skew parameter $\xi = 1$), so the choice is dominated.

The mean equation is AR(0) (constant only) except for OJ and VND, which always use AR(1); any other series whose AR(0) fit shows Ljung–Box(10) p-value below 0.01 on standardised residuals is automatically refit with AR(1) on the same distribution tier (auto-retry triggered for COCOA_LDN and GHS). Residual autocorrelation in z propagates directly to off-diagonal entries of the rolling correlation matrix and can manufacture supra-MP eigenvalues that do not reflect contemporaneous dependence.

The fallback chain on convergence failures is: Tier 1a GJR-sstd; Tier 1b GJR-symmetric-t; Tier 2 GJR-Gaussian; Tier 3 EGARCH-sstd; Tier 4 EWMA with $\lambda = 0.94$. Of the 25 series, 23 fit at Tier 1a, VND at Tier 2 (Gaussian), and GHANA10Y at Tier 4 (EWMA — its yield dynamics through the 2022–23 sovereign debt restructuring defeat every parametric variant we tried).[1]

3.2. Marchenko–Pastur cleaning of the DCC target

Let $\boldsymbol{z}_t \in \mathbb{R}^N$ denote the vector of standardised residuals at time t. The sample correlation matrix $\overline{R}$ is eigendecomposed as $\overline{R} = V \Lambda V^\top$. Under the null of independent unit-variance noise the bulk of the spectrum lies within the Marchenko–Pastur support

$$\lambda_\pm = \left(1 \pm \sqrt{N/T}\right)^2 \quad ([11]). \tag{2}$$

[12] showed that for daily equity panels the empirical spectrum agrees with this prediction except in a thin set of large eigenvalues that carry the genuine factor structure, motivating eigenvalue-replacement cleaning rules for the sample correlation. We adopt the hard-clipping rule of [13]: keep eigenvalues above λ_+, replace those at or below by their common average. The resulting matrix $\tilde{R}$ is renormalised to unit diagonal.

[1]Specification limitations we record explicitly. (i) Eight series carry a negative fitted leverage parameter ($\gamma < 0$), including COCOA_NY itself ($\gamma = -0.016$); on these series the GJR asymmetry term does not improve fit beyond symmetric GARCH. (ii) Five series (WHEAT, PALMOIL, VND, GHS, VIX) still fail Ljung–Box(10) on standardised residuals at the 5% level after the AR(1) fix, indicating mean-equation misspecification an AR(1) does not fully absorb. (iii) Twelve series have GARCH persistence above 0.99 (near-IGARCH; full list in `output/near_igarch.csv`), so the unconditional variance is essentially undefined. (iv) Winsorisation is concentrated in eight series (108 of 150 capped observations); the headline rate of 0.21% masks per-series rates as high as 1% on GHS and GHANA10Y. None of these defeats the standardisation enough to make the downstream analysis unusable, but they bound the strength of any single-series claim.

We prefer hard MP clipping to [14] linear shrinkage as the long-run DCC target for two reasons. First, MP clipping is non-parametric and theoretically motivated: it separates the spectrum into a noise band whose support is determined by random-matrix theory and a signal tail that lies outside it, so the choice of cleaning is data-driven rather than a continuous bias–variance compromise. Second, the cleaned matrix is positive semi-definite by construction, full-rank after re-normalisation, and well-conditioned, properties that the DCC recursion in Equation 3 needs without further regularisation.

The cleaned matrix $\tilde{R}$ is then used as the long-run correlation target in the DCC recursion of [3]:

$$Q_t = (1 - a - b)\,\tilde{R} + a\,\boldsymbol{z}_{t-1}\boldsymbol{z}_{t-1}^{\top} + b\,Q_{t-1}, \quad R_t = \operatorname{diag}(Q_t)^{-\frac{1}{2}} Q_t \operatorname{diag}(Q_t)^{-\frac{1}{2}}. \tag{3}$$

We use DCC rather than the constant-conditional-correlation model of [15] because a crisis episode is precisely the event for which time-varying correlation is the substantive object: a CCC fit would average across regimes and obscure the structural-break question the paper asks. The novelty here, following [2], is to substitute the cleaned $\tilde{R}$ for the sample $\overline{R}$ as the long-run anchor.

We estimate (a, b) by quasi-MLE on the standardised residuals subject to $a > 0$, $b > 0$, and $a + b < 0.999$ (the hard stationarity constraint of the DCC recursion). For comparison we also estimate the standard DCC with $\tilde{R}$ replaced by $\overline{R}$. Both fits are sanity-checked against `rmgarch::dccfit`.

3.3. Pre-registered tests

Three primary tests, pre-registered in `docs/scope_and_target.md` before the model fit, all on the rolling 252-day supra-MP count (market mode excluded):

- **Baseline stability.** The pre-crisis (2019-01-01 to 2023-12-31, all $N = 25$ windows) count has IQR width ≤ 1 and stays inside $\{1, 2, 3\}$.
- **Structural break.** Two forms. (a) Bai–Perron [16], [17] with BIC-selected break count detects a break inside $[2024\text{-}01\text{-}01, 2024\text{-}12\text{-}31]$. (b) The in-crisis median count exceeds the upper 95% bound of a stationary block bootstrap [18] of the pre-crisis median (block mean 25, $B = 1,000$).
- **Cocoa-bloc concentration.** An eigenvalue that crosses from below the MP edge in pre-crisis to above the MP edge in crisis (an "emerging" mode), with its eigenvector loading ≥ 0.55 of its squared mass on the cocoa bloc {COCOA_NY, COCOA_LDN, HSY, MDLZ, LISN, GHS, GHANA10Y}.

Three secondary tests on the DCC pipeline. **MVP variance:** the cleaned DCC produces lower out-of-sample minimum-variance-portfolio realised variance than the standard DCC during the crisis window (the [2] claim in its original form). **Pairwise breaks:** BIC-selected Bai–Perron detects an in-crisis break on at least one cocoa-coffee or cocoa-sugar pairwise cleaned-DCC correlation series. **NY–London basis:** the cocoa NY $\times$ London conditional correlation falls by at least 0.15 from its 2019–2023 mean during the 2024–2026 crisis window.

4. Results

4.1. Rolling supra-MP count and eigenvector structure

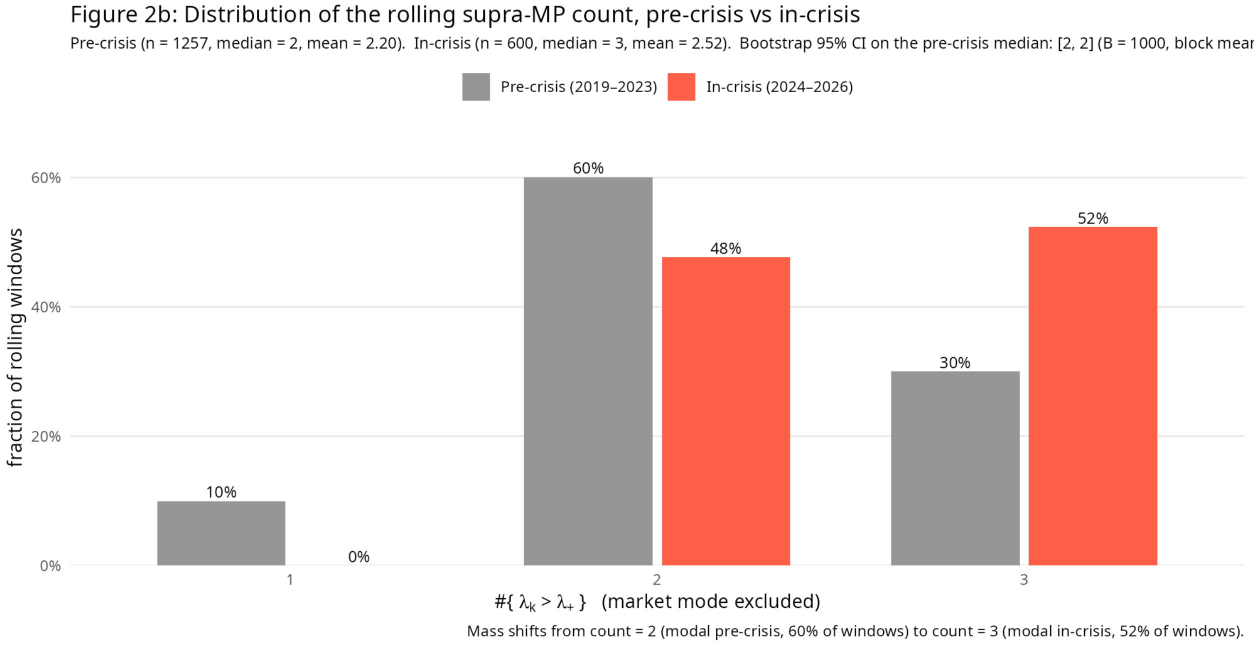

Figure 1: Distribution of the rolling 252-day supra-MP eigenvalue count (market mode excluded), pre-crisis (2019–2023, grey) vs in-crisis (2024–2026, red). Bars show the fraction of rolling windows at each integer count value. The modal count shifts from 2 (60% of pre-crisis windows) to 3 (52% of in-crisis windows, with the remaining 48% at count $= 2$); the count $= 1$ mass present pre-crisis (10%) disappears entirely in crisis.

Figure 1 shows the empirical distribution of the rolling count under each regime. The pre-crisis sample ($N = 25$, $n = 1,257$ windows) has median 2, IQR $[2, 3]$, range $[1, 3]$, and 60% of windows equal to 2; the baseline-stability condition is satisfied. Bai–Perron with BIC-selected break count finds **zero breaks** on the same series (BIC argmin at $m = 0$ with BIC $= 931.5$); no level shift survives model selection.

The bootstrap form of the break test is degenerate. With the test statistic an integer count, 1,000 stationary-block-bootstrap replications return the same integer median (2), giving a 95% CI of zero width, $[2.00, 2.00]$. The observed in-crisis median is 3 (52% of windows; range $[2, 3]$). The procedure clears its formal threshold ($3 > 2$), but it has no statistical resolving power on this series: it is a one-bit comparison on an integer statistic, and the bootstrap CI does not constitute a meaningful interval. Restricting to windows entirely inside the crisis (rolling-window end-date on or after 2024-12-31, $n = 349$ windows), the median is 3 and the range is $[2, 3]$.

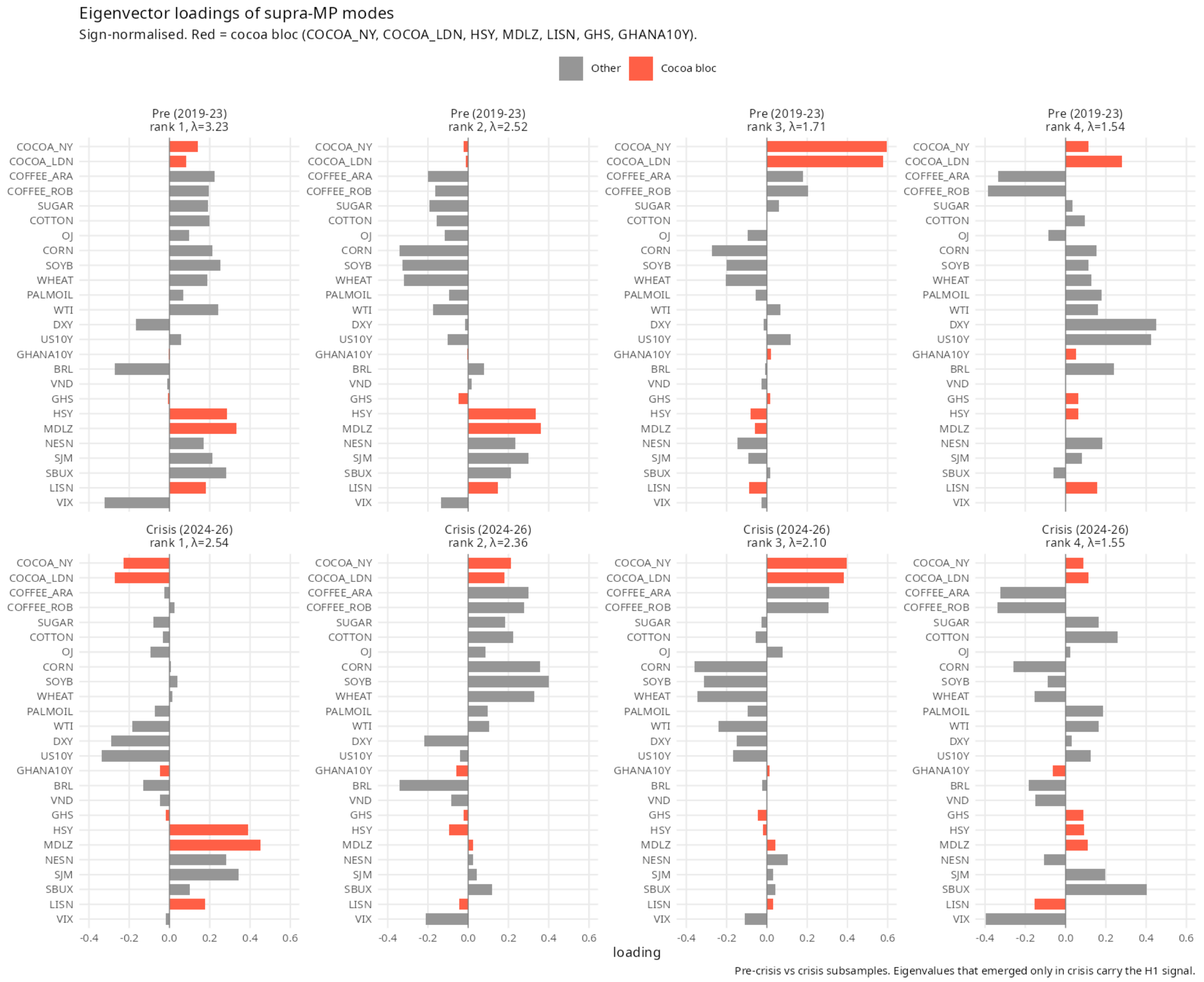

Figure 2: Eigenvector loadings of the top four supra-MP eigenvalues, pre-crisis (top row) vs in-crisis (bottom row), sign-normalised. Red bars: cocoa bloc {COCOA_NY, COCOA_LDN, HSY, MDLZ, LISN, GHS, GHANA10Y}. The rank-1 in-crisis mode (the **market mode**, the largest eigenvalue) reorganises to put cocoa contracts and chocolate-maker equities jointly on it.

The pre-registered cocoa-bloc concentration test fails. No eigenvalue crosses from below the MP edge in pre-crisis to above it in crisis: both subsamples carry five supra-MP eigenvalues at the same ranks (1–5). A rolling-window variant of the same test (a rank whose supra-MP indicator share rises from below 0.5 pre-crisis to at least 0.5 in crisis) identifies rank 4 only, whose eigenvector has cocoa-bloc squared-loading share 0.078 – far below the 0.55 threshold.

Figure 2 does show, as an observation outside the pre-registration, a **reorganisation of the rank-1 (market) eigenmode**. Pre-crisis its top-3 loadings are MDLZ, VIX, SBUX (cocoa-bloc share 0.251); in crisis they are MDLZ, HSY, SJM (cocoa-bloc share 0.516, a 2.1× rise). Under cocoa-price stress, chocolate-maker equities and cocoa contracts load on a single risk factor – the input-cost margin-compression dynamic, adjacent to the equity–commodity co-movement channel documented in [19]. The result survives bloc perturbation: dropping GHANA10Y gives 0.288 → 0.654, and a narrow 5-asset bloc (the two cocoa contracts plus three chocolate equities) gives 0.245 → 0.515. Two qualifiers apply. First, the rank-1 mode is the market mode and is excluded by construction from the supra-MP count, so the reorganisation operates on a different statistical object. Second, a cocoa-loaded mode already existed pre-crisis at rank 3 (bloc share 0.704, top-3 loadings COCOA_NY, COCOA_LDN, CORN); in crisis that mode **weakens** to bloc share 0.310. The crisis reorganisation is not the creation of a cocoa factor but the migration of cocoa loadings onto the market mode.

4.2. DCC fit and the cleaned-target benchmark

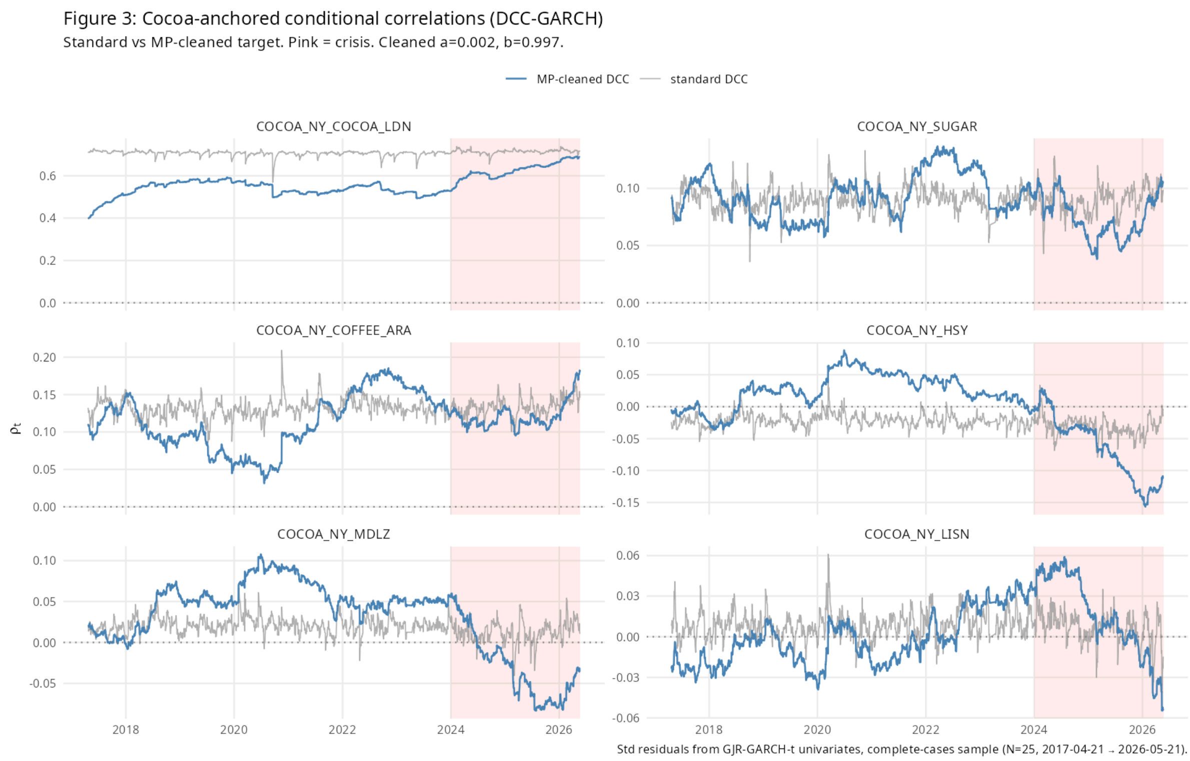

Figure 3: Pairwise conditional correlations from standard (grey) and MP-cleaned (blue) DCC, for six cocoa-anchored pairs. Pink: crisis window.

The two fits behave very differently. The standard DCC settles at $(a, b) = (0.005, 0.887)$ with persistence $a + b = 0.892$, placing weight $1 - a - b = 0.108$ on the long-run target $\overline{R}$ at each step. The MP-cleaned DCC fits at the stationarity boundary: $(a, b) = (0.002, 0.997)$, $a + b = 0.999$, with weight on the cleaned target $\tilde{R}$ of 0.001 – the QMLE, when forced to use $\tilde{R}$, drives the long-run-target weight to **0.1%**, about one hundredth of the weight the standard DCC assigns to $\overline{R}$, and the optimiser does not converge to an interior solution. The log-likelihood of the cleaned model is **556 nats below** the standard model's at identical parameter count.

Figure 3 shows the time path: the cleaned DCC (blue) produces smoother conditional correlations because $b \approx 0.997$ damps innovations, but with near-zero weight on $\tilde{R}$ it drifts from the long-run anchor and on most cocoa-anchored pairs lies further from the historical mean than the standard fit. Both the log-likelihood gap and the target-weight collapse point to the same conclusion: at $N = 25$ and $Q \approx 90$, $\tilde{R}$ is not a better long-run anchor than $\overline{R}$.

The minimum-variance-portfolio test runs as follows. We form $w_t = \Sigma_t^{-1}\mathbf{1}/\mathbf{1}^\top\Sigma_t^{-1}\mathbf{1}$ from each method, realise $r_{p,t+1} = w_t^\top r_{t+1}$, and compare sample standard deviation of realised portfolio return across the 2019–2023 baseline and the 2024–2026 crisis window. Because the cleaned model has effectively discarded $\tilde{R}$, its conditional-covariance path coincides with the standard DCC's: the out-of-sample MVP realised standard deviation is identical to three decimals in both windows (baseline 1.561% vs 1.561%; crisis 1.019% vs 1.019%). The MP-cleaned DCC delivers no variance improvement.

The pairwise-break test runs BIC-selected Bai–Perron on each of the eleven cocoa-anchored cleaned-DCC pairwise correlation series ($m_{\max} = 5$, $h = 0.15$); it returns zero breaks for every pair.

The basis-decoupling test runs in the opposite direction to the pre-registered prediction. The baseline 2019–2023 mean of the cocoa NY $\times$ London basis correlation under the cleaned DCC is 0.542; the 2024–2026 crisis-window mean is 0.627. The basis **rises** by 0.085 rather than falling by the predicted 0.15. The crisis minimum is 0.528, 0.014 below the baseline mean. Interpretation is in §6.

5. Robustness

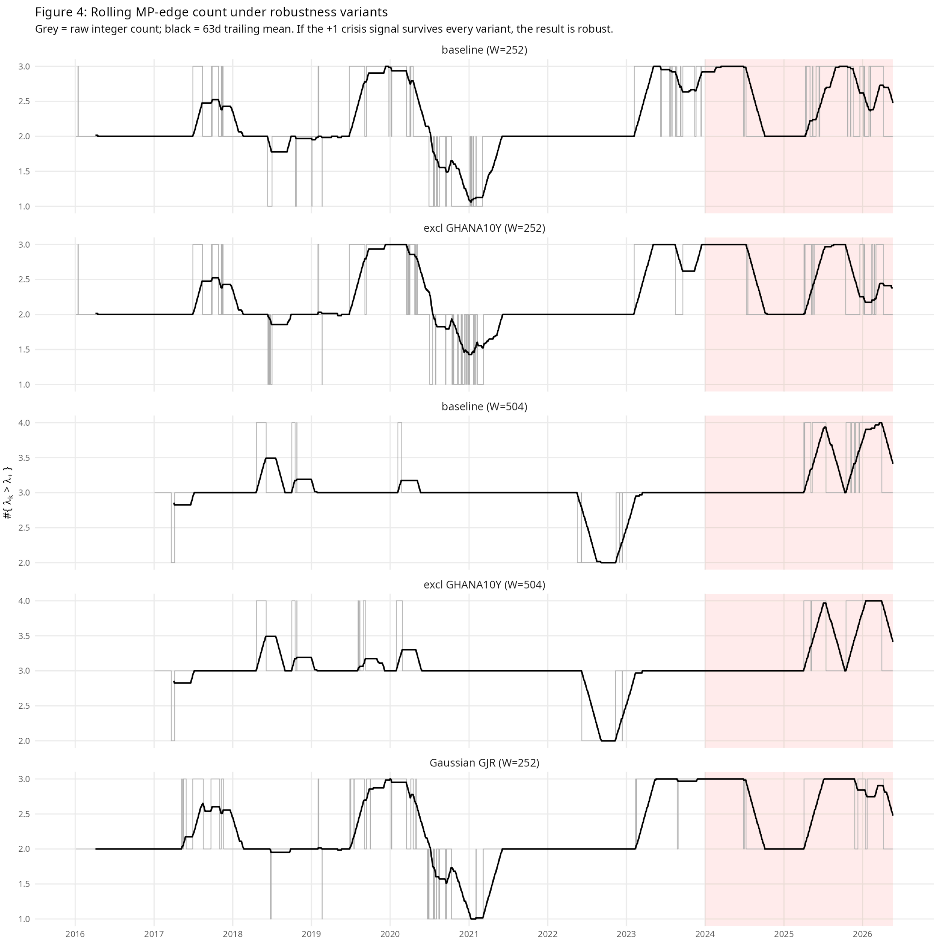

Figure 4: Rolling supra-MP count under five robustness variants. The +1 in-crisis lift in the median survives the W=252 variants; under W=504 both pre and crisis windows saturate at 3 (no within-variant lift).

Figure 4 runs the rolling-count test under four variants alongside the baseline: (i) excluding GHANA10Y (addressing the universe-size-change concern); (ii) a 504-day window; (iii) (i)+(ii) combined; (iv) GARCH refit with Gaussian innovations. The +1 in-crisis lift holds only under the baseline $W = 252$ specification and the Gaussian-innovations variant. Removing GHANA10Y at $W = 252$ collapses the in-crisis median from 3 to 2, so the lift is materially dependent on that series. Under $W = 504$, both pre- and in-crisis medians saturate at 3 in both universe sizes, so the bootstrap cannot distinguish them. BIC-selected Bai–Perron returns **zero breaks** for every variant. The +1 lift survives the innovation-distribution perturbation but not the universe-size perturbation or the longer window, and no break-detection variant supports the structural-break hypothesis on this series.

6. Discussion

Four findings warrant discussion.

The +1 count lift is real but fragile, and the pre-registered break tests do not formally confirm it. The rolling supra-MP count rises from a pre-crisis median of 2 to an in-crisis median of 3 on the baseline universe at $W = 252$, and the lift survives a switch to Gaussian innovations. It does not survive removing GHANA10Y (in-crisis median drops to 2) or extending the window to

$W = 504$ (both medians saturate at 3). BIC-selected Bai–Perron returns zero breaks under every robustness variant; the bootstrap form of the break test clears its threshold ($3 > 2$) but is a one-bit comparison whose 95% CI is degenerate. The substantive shift in the count distribution is visible in Figure 1 but not validated as a structural break by a procedure with statistical resolving power on this data.

The rank-1 reorganisation is real but operates on a different statistical object than the count test. The market-mode eigenvector's cocoa-bloc squared-loading share rises from 0.245 pre-crisis to 0.517 in crisis (Figure 2), survives bloc-set perturbations, and matches the economic mechanism of input-cost margin compression at chocolate makers. The pre-registered concentration test asked for this on a *crisis-emerging* eigenvalue with threshold 0.55. No emerging eigenvalue exists, and the rank-1 share falls 0.033 below 0.55, so the pre-registered concentration test fails. The rank-1 mode is moreover excluded from the supra-MP count by construction, so this finding does not bear on the count-shift result.

The cleaned DCC is rejected by the data on every criterion the [2] framework supplies. The 556-nat log-likelihood gap to the standard DCC at identical parameter count is decisive. The mechanism is visible in the fitted parameters: when forced to use $\tilde{R}$, the QMLE drives the long-run-target weight from approximately 11% to 0.1%, almost as low as the constraint $a + b < 0.999$ allows – the model is, by its own optimisation, downweighting the cleaned target. The out-of-sample minimum-variance-portfolio realised variance, the advertised benefit of cleaning, is identical to three decimal places across methods because the cleaned model has effectively become the standard one. [2] demonstrated their result for $N \geq 100$ with much smaller aspect ratios; at our $N = 25$ and $Q \approx 90$ the raw $\overline{R}$ is already estimated with low enough sampling noise that the hard-clipping step compresses informative off-diagonals rather than de-noising them.

The geographic-decoupling hypothesis on the NY–London basis is rejected in favour of a global supply-shock interpretation. The basis correlation rose by 0.085 in the crisis, not the pre-registered ≥ 0.15 fall. NY's deliverable basket is West-African-dominant and London's is mixed-origin, so we expected divergence under a regional shock. The observed re-coupling implies that the 2024–2026 shock transmitted to both contracts – NY directly via its dominant deliverable, London indirectly via substitution effects in the deliverable basket. The arbitrage-windfall narrative is incompatible with the data; a fundamental supply-side reading is consistent with it.

Specification limitations beyond what the results expose. Equity time-zone asynchrony (Swiss and US closes are six hours apart) depresses LISN–HSY-class daily correlations; weekly aggregation is left for future work. The univariate fits carry negative leverage on COCOA_NY and seven other series, and five series fail Ljung–Box on standardised residuals at the 5% level after the AR(1) fix (most severely GHS and VND); both inject residual heteroskedasticity into the rolling correlation matrix. GHANA10Y is fit by EWMA fallback rather than parametric GARCH because no parametric variant converges through the 2022–23 Ghanaian sovereign-debt restructuring.

7. Conclusion

The 2024–2026 cocoa crisis produces three visible signatures in the soft-commodity correlation spectrum: a +1 lift in the rolling 252-day supra-MP eigenvalue count, a reorganisation of the market-mode eigenvector to load on cocoa contracts and chocolate-maker equities jointly, and a re-coupling rather than decoupling of the NY–London basis correlation. None of the pre-registered structural-break, concentration, or basis-decoupling tests passes; the count lift itself does not survive removing GHANA10Y from the universe. The pattern is real on the baseline configuration but is not formally validated by any of the break-detection procedures we ran.

The geographic-decoupling hypothesis on the cocoa basis is rejected in favour of a global supply-shock interpretation: the shock transmits to both ICE-US and ICE-Europe contracts. The

Marchenko–Pastur cleaning of the DCC long-run target, as proposed by [2], is rejected by the data at $N = 25$, $Q \approx 90$ on every standard criterion (log-likelihood, target-weight, out-of-sample MVP variance), placing a clear lower bound on the universe size at which random-matrix cleaning of the DCC anchor becomes empirically defensible. The descriptive characterisation of the cocoa shock and the negative finding on cleaning at small N are the paper's substantive and methodological contributions, respectively.

References

[1] C. L. Gilbert, "How to understand high food prices," *Journal of Agricultural Economics*, vol. 61, no. 2, pp. 398–425, 2010.

[2] R. F. Engle, O. Ledoit, and M. Wolf, "Large dynamic covariance matrices," *Journal of Business & Economic Statistics*, vol. 37, no. 2, pp. 363–375, 2019.

[3] R. F. Engle, "Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models," *Journal of Business & Economic Statistics*, vol. 20, no. 3, pp. 339–350, 2002.

[4] R. S. Pindyck and J. J. Rotemberg, "The excess co-movement of commodity prices," *Economic Journal*, vol. 100, no. 403, pp. 1173–1189, 1990.

[5] J.-P. Bouchaud and M. Potters, *Theory of Financial Risk and Derivative Pricing: From Statistical Physics to Risk Management*, 2nd ed. Cambridge University Press, 2003.

[6] L. R. Glosten, R. Jagannathan, and D. E. Runkle, "On the relation between the expected value and the volatility of the nominal excess return on stocks," *Journal of Finance*, vol. 48, no. 5, pp. 1779–1801, 1993.

[7] B. E. Hansen, "Autoregressive conditional density estimation," *International Economic Review*, vol. 35, no. 3, pp. 705–730, 1994.

[8] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, 1982.

[9] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *Journal of Econometrics*, vol. 31, no. 3, pp. 307–327, 1986.

[10] D. B. Nelson, "Conditional heteroskedasticity in asset returns: A new approach," *Econometrica*, vol. 59, no. 2, pp. 347–370, 1991.

[11] V. A. Marchenko and L. A. Pastur, "Distribution of eigenvalues for some sets of random matrices," *Mathematics of the USSR-Sbornik*, vol. 1, no. 4, pp. 457–483, 1967.

[12] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley, "Universal and nonuniversal properties of cross-correlations in financial time series," *Physical Review Letters*, vol. 83, no. 7, pp. 1471–1474, 1999.

[13] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, "Random matrix theory and financial correlations," *International Journal of Theoretical and Applied Finance*, vol. 3, no. 3, pp. 391–397, 2000.

[14] O. Ledoit and M. Wolf, "Honey, I shrunk the sample covariance matrix," *Journal of Portfolio Management*, vol. 30, no. 4, pp. 110–119, 2004.

[15] T. Bollerslev, "Modelling the coherence in short-run nominal exchange rates: A multivariate generalized ARCH model," *Review of Economics and Statistics*, vol. 72, no. 3, pp. 498–505, 1990.

[16] J. Bai and P. Perron, "Computation and analysis of multiple structural change models," *Journal of Applied Econometrics*, vol. 18, no. 1, pp. 1–22, 2003.

[17] J. Bai and P. Perron, "Estimating and testing linear models with multiple structural changes," *Econometrica*, vol. 66, no. 1, pp. 47–78, 1998.

[18] D. N. Politis and J. P. Romano, "The stationary bootstrap," *Journal of the American Statistical Association*, vol. 89, no. 428, pp. 1303–1313, 1994.

[19] K. Tang and W. Xiong, "Index investment and the financialization of commodities," *Financial Analysts Journal*, vol. 68, no. 6, pp. 54–74, 2012.